Def. Let M be an oriented regular surface. A map

$$N: M \to S^2; p \mapsto n(p)$$

is called the Gauss map.

Since $N(p)$ is perpendicular to T_p and also perpendicular to $T_{N(p)}S^2$, therefore T_pM and T_pS^2 are parallel.

Let $x \in T_pM$. Choose a smooth curve

$$\alpha: (-\varepsilon, \varepsilon) \to M \text{ such that } \alpha(0) = p \text{ and } \alpha'(0) = x.$$

Now, define $dN_p(x) = \left.\frac{d}{dt}\right|_{t=0} N(\alpha(t))$

As usual convention, define the shape operator is

$$S_p: T_pM \to T_pM: x \mapsto -dN_p(x).$$

Consider a patch $X: U \to M$ s.t. $X(u_0, v_0) = p$, and $x = aX_u + bX_v$.

$$dN_p(x) = aN_u + bN_v.$$